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A chemo-thermo-mechanical coupled phase field framework for failure in thermal barrier coatings

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Abstract

The failure process in a thermal barrier coating (TBC) system can be complicated due to complex microstructure, material property mismatch, high temperature change, chemical reactions during the failure process, etc. It brings considerable difficulties to the modeling technique. A complete failure process in TBC contains multiple failure types, e.g., bulk fracture, delamination of TC (top coating)/TGO (thermally grown oxide) interface and TGO/BC (bond coating) interface, and the interaction among them. However, existing approaches mainly focus on a single failure type and thus are faced with limits on exploring the complete failure mechanism. In this work, we propose a chemo-thermo-mechanical coupled phase field model based on thermodynamic laws, such that the complete TBC fracture process can be captured. The present contribution provides a step forward in comparison with the existing phase field models. Couplings among thermal conduction, material oxidation and damage are considered, and the fundamental differential equations are derived strictly follow the thermodynamic laws. The proposed model is first validated through several benchmarks, and then followed by several simulations of the failure process of a TBC system. The simulations and experiments show that the bulk fracture occurs when the TGO is thin at the cooling process, the growth effect of TGO layer causes damage initiation at the interface and finally transformed into the interfacial delamination, the interaction of interface cracking and bulk material cracking is captured. The proposed model is proven to be a powerful tool in solving the failure process of a TBC system and is promising to handle other similar problems.
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Keywords: Phase field model; Oxidization; Thermal barrier coating; Delamination

1. Introduction

Thermal barrier coatings (TBCs) are designed to insulate and protect the super alloy substrate from extreme high temperature. They are usually used to protect the metallic parts used in aircraft engines, gas turbines, or power

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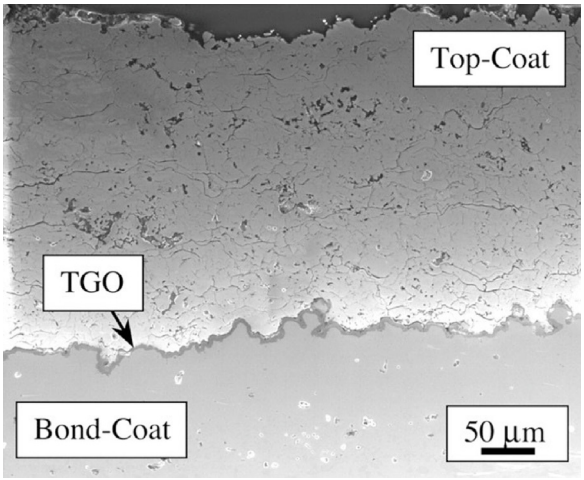


Fig. 1. Cross-sectional SEM of an air-plasma-sprayed (APS) TBC [2].

generation applications [1]. A TBC system in general consists of three layers of different materials (Fig. 1), they are: (a) thermal insulation ceramic coat layer in the top (TC), (b) metallic bond coat (BC) layer in the bottom, and (c) thermally grown oxide (TGO) layer in between. The ceramic coating material can be deposited on a metallic substrate either by means of air-plasma-spray (APS) deposition or electron beam physical-vapor deposition (EB-PVD). Microstructure and interfacial morphology of ceramic coatings are different of these two kinds, and we will limit the scope to APS coatings in this work.

The failure mechanism of a TBC system can be extremely complicated, the most important factors that contribute to the complexity are (a) mismatch of coefficients of thermal expansion (CTEs) among different materials; (b) swelling of TGO layer during oxidation process of BC material; (c) micro-defects (micro-cracks, pores and voids) and (d) the complex microstructure of the interface between TGO and TC. The stress can be considerably high in the vicinity of the interface, and the reason could be twofold: the mismatch of CTEs under the cooling process leads to thermal stresses and the swelling of TGO growth at an elevated temperature leads to the material and stress redistribution [3]. Upon the cooling process, when the TGO layer is thin or absent, the thermal mismatch leads to a tensile stress state at the summit of the TC/TGO interface and a compression stress state at the valley. When the TGO layer becoming thicker, the tensile area may shift to the valley of the interface and the compression area to the summit [4].

To correctly predict the growth of the TGO layer during which the BC material gradually transforms into the TGO material, resulting in the volumetric expansion and change in material properties. A most common and simple approach is to adopt an empirical formula to approximate the relation between the thickness of the TGO layer and time, which is determined experimentally for different cases [5,6]. Another approach is to consider the growth process of the TGO layer through a diffusion and reaction equation [7–9].

The failure process in TBC is rather complicated and the analysis relies heavily on the numerical methods. Early works use thermo-mechanical finite element method (FEM) in combination with stress based failure criteria to study the effect of the residual stress [10–12]. These modeling techniques can successfully predict crack initiation timing and position, but can hardly be used to predict crack propagation which is important in understanding the failure mechanism of a TBC system. Other methods such as virtual crack closure technique (VCCT), extended FEM (XFEM) and cohesive zone method (CZM) are also employed. Bumgardner et al. [13] studied the mechanism of delamination and vertical crack propagation by using in-situ high-temperature digital image correlation (DIC) in combination with VCCT. Other researchers [6,14] used the XFEM to capture the TC and TGO cracking processes, and CZM to capture the interfacial cracking process. To simulate the TC fracture with cohesive elements, Krishnasamy et al. [15] inserted cohesive elements throughout the inter-element boundaries which enables the crack to propagate along an arbitrary direction and path. However, the method is also considered to be mesh dependent. Wang et al. [16] used the peridynamic method to investigate the cooling-induced cracking process of the TC layer. Zhou et al. [8] used a phase field method to simulate the TGO cracking process and cohesive elements to simulate

the TC/TGO interfacial delamination, but the interaction between the two failure types was not considered hence the complete failure process cannot be captured which may be less satisfactory in understanding the failure mechanism.

The phase field method has drawn attentions in recent years, it was originated and developed independently from two communities: (a) Landau and Ginzburg formalism in physics community, and (b) the Francfort–Marigo variational formulation of brittle fracture [17] and its regularized version by Bourdin et al. [18,19] in mechanics community [20]. Phase field model has been applied for various problems, such as brittle fracture [21–27], quasi-brittle fracture [28–35], ductile fracture [36–43], dynamic fracture [44–51] and cohesive fracture [52–57], and materials with different properties and microstructures, such as isotropic material [44,51,58], anisotropic material [59–62], porous media [63–65], heterogeneous material [66,67], etc. Phase field model is a smeared crack model, which represents the crack topology using a continuous distributed scalar field, *i.e.*, the phase field. Unlike the discrete crack model (XFEM for example) which defines the displacement jump across the crack surfaces explicitly, the displacement is continuous across the crack in phase field model. Phase field model can predict fracture initiation, propagation, coalescence and branching within one framework, it has eliminated remeshing, tracking the crack trajectory, and the necessity of dealing with the singular stress around the crack tip area which could be troublesome for existing methods. Solving fracture problems using the phase field model can be easily done by solving a system of equation of a multi-field problem and the theory can be implemented into commercial finite element packages to have higher solving efficiency and capability (such as ABAQUS [22,23,68]).

We propose in this paper a chemo-thermo-mechanical coupled phase field model to simulate the complete fracture process of a TBC system. Both the interfacial delamination and the bulk crack process are considered in a unified framework. The couplings among thermal conduction, oxidization, and damage are fully considered. Plastic deformation, high temperature creep effect, and sintering are not considered in this work for simplicity (given that they are minor factor in the failure mechanism studied in this work compared with TGO swelling and mismatch of CTEs [8,13,14,16]). As the crack topology is represented by phase field in a smeared manner, the interface delamination cannot be simulated directly, one can combine a phase field model with interface element model, e.g., cohesive zone model [69], or choose to represent the interface in a smeared manner [70–72] with a modified fracture toughness. We adopted the latter approach in this work and proposed a scalar field variable, *i.e.* interface phase field, to smear the interface properties and keep the material properties change continuously.

This article is organized as follows. We first consider in Section 2 the basic thermodynamic laws for the continuum body, then the governing equations for each physical field are obtained, the interface treatment is introduced and the modified fracture toughness is obtained using dissipation equivalence. In Section 3 we formulate the finite element discretization for each field. A parametric study is given in Section 4. We provide in Section 5 a few numerical examples to validate the proposed phase field model.

2. Thermodynamically consistent phase field model for chemo-thermo-mechanical coupled problem

2.1. Phase field model for fracture

To introduce phase field model, we assume a region $\Omega \subset \mathbb{R}^{\dim}(\dim= 2, 3)$ of the solid body with external boundary $\partial\Omega = [\partial\Omega_u \cup \partial\Omega_t] \subset \mathbb{R}^{\dim-1}$ and internal discontinuity boundary $\Gamma_c \subset \mathbb{R}^{\dim-1}$ at time t (consider only mechanical boundaries for simplicity), as shown in Fig. 2a. The external boundary is partitioned into two disjoint parts $\partial\Omega_u$ and $\partial\Omega_t$, so that $\partial\Omega_u \cap \partial\Omega_t = \varnothing$ and $\partial\Omega_u \cup \partial\Omega_t = \partial\Omega$. The displacement of a point $\boldsymbol{x}$ at time t is denoted by $\boldsymbol{u}: \Omega \rightarrow \mathbb{R}^{\dim}$. Prescribed displacement $\overline{\boldsymbol{u}}$ and traction $\boldsymbol{t}$ are acting on $\partial\Omega_u$ and $\partial\Omega_t$, respectively. The closure of the region is $\overline{\Omega} \equiv \Omega \cup \partial\Omega \cup \Gamma_c$.

In the discrete fracture model context, the crack's geometry is explicitly represented, which can be achieved by either using a conforming mesh or by adding enrichment functions to the solution space. Phase field model, as a smeared method, approximates the crack topology with a damage zone represented by a scalar field variable $d \in [0, 1]$ (Fig. 2a). The value of d can be deemed as a measure of damage state of the material, *i.e.*, the material is intact when $d = 0$, and fully failed when $d = 1$. A scalar internal length scale parameter l is used to control the width of the regularization area. We define the regularized crack surface density functional as [52]

$$\Gamma_l(d) := \frac{1}{c_0 l}\int_{\Omega}\left[\alpha(d) + l^2|\nabla d|^2\right]\mathrm{d}V \equiv \int_{\Omega}\gamma(d, \nabla d)\,\mathrm{d}V \tag{1}$$

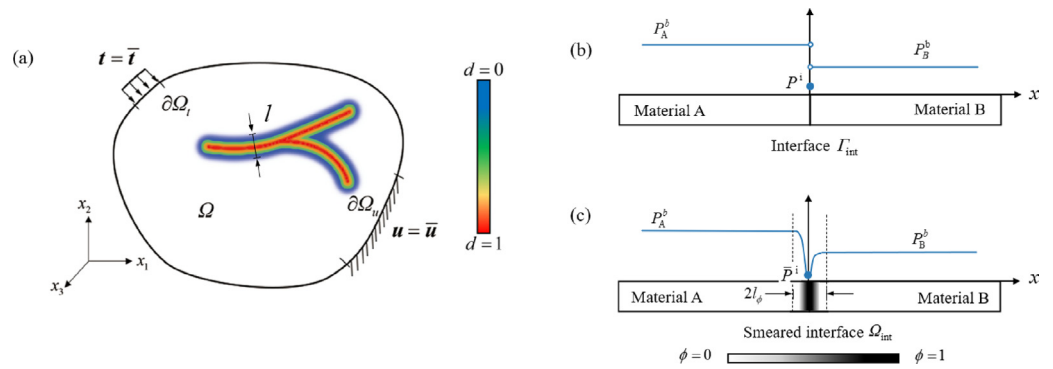


Fig. 2. (a) Configuration of a solid body containing a smeared crack topology represented by phase field $d(\boldsymbol{x}, t)$, the width of the smeared damage zone is controlled by an internal length scale parameter l. (b) Schematic of a smeared interface property P (fracture toughness or strength in this work), P^{b} is the property of the bulk material, P^{i} is the property of the interface, the material property has been smeared by using an interface phase field ϕ into a region Ω_{int}, the bandwidth of the smeared interface is $2l_\phi$.

where $\alpha(d)$ is the crack geometric function which characterizes homogeneous evolution of phase field, $c_0 = 4\int_0^1 \sqrt{\alpha(x)}\,\mathrm{d}x$ is a scaling parameter ensuring the regularized crack surface approximates the discrete crack surface Γ_c, $\gamma(d, \nabla d)$ is the crack surface density functional.

The crack geometry adopted in this work has the following form [52]

$$\alpha(d) = 2d - d^2 \tag{2}$$

And the corresponding scaling parameter c_0, half-bandwidth of the regularization area b and the profile of the phase field $d(x)$ can be obtained

$$c_0 = \pi, \quad b = \frac{\pi}{2} l \quad \text{and} \quad d(x) = 1 - \sin\left(\frac{|x|}{l}\right) \tag{3}$$

2.2. Smeared interface

To introduce the interface phase field method, we first consider a 1D bar with an interface Γ_{int} as shown in Fig. 2b, the fracture toughness of the interface is G_i. We introduce an interface phase field variable $\phi \in (0, 1)$ to smear the interface properties over a zone Ω_{int} with a half-bandwidth of l_ϕ (Fig. 2c), the value of ϕ continuously changes from 1 at the middle of the smeared interface to 0 when the distance is equal to l_ϕ. The fracture toughness and strength (on the side of material A for instance) can be written as

$$\begin{cases} G_c(x) = \left[1 - (1 - \phi(x))^2\right] G_c^i + (1 - \phi(x))^2 G_c^{\mathrm{A}} \\ f_t(x) = \left[1 - (1 - \phi(x))^2\right] f_t^i + (1 - \phi(x))^2 f_t^{\mathrm{A}} \end{cases} \tag{4}$$

where G_c^{A} and G_c^i are the fracture toughness of the bulk material and the interface, respectively, f_t^i and f_t^{A} are the interface strength and bulk material strength, respectively.

According to [70,71], a compensation on the equivalent material property should be introduced to guarantee the dissipation equivalence. We have the following relation under the 1D setting

$$G_i = \int_{-b}^{b} \left\{ \left[1 - (1 - \phi)^2\right] \overline{G}_c^i + (1 - \phi)^2 G_c^{\mathrm{A}} \right\} \gamma \, \mathrm{d}x \tag{5}$$

where $\overline{G}_c^i$ is the modified fracture toughness of the interface, the dissipation equivalence ensures the energy release rate of the smeared interface equal to G_c^i (as illustrated in Fig. 3a), and $\overline{G}_c^i$ can be obtained by solving Eq. (5) as follows

$$\overline{G}_c^i = \frac{G_i - G_c^{\mathrm{A}} \int_{-b}^{b} (1 - \phi)^2 \gamma \, \mathrm{d}x}{\int_{-b}^{b} \left[1 - (1 - \phi)^2\right] \gamma \, \mathrm{d}x} \tag{6}$$

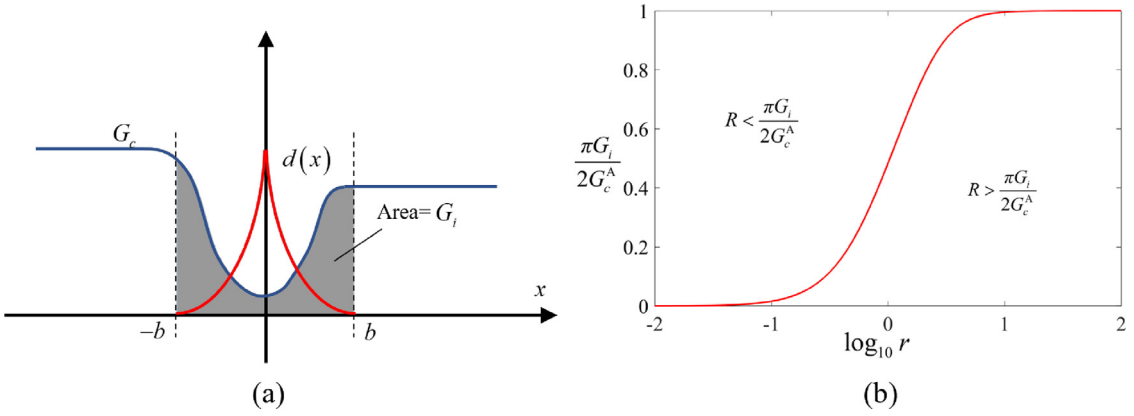


Fig. 3. (a) Schematic illustration of dissipation equivalence, the integral of $G_c(x)$ from $-b$ to b equals to G_i (b) $\pi G_i/2G_c^{\mathrm{A}}$ vs. $\log_{10} r$, where $r = l/l_\phi$.

We choose the following exponential function as the profile of the interface phase field

$$\phi = e^{-\frac{|x|}{l_\phi}} \tag{7}$$

To avoid the non-physical negative value of $\overline{G}_c^i$, the following constraint must be satisfied

$$\int_{-b}^{b} (1-\phi)^2 \gamma \, \mathrm{d}x < \frac{G_i}{G_c^{\mathrm{A}}} \tag{8}$$

Using the expressions Eqs. (1), $(3)_3$ and (7), Eq. (8) can be rewritten as

$$R(r) = \frac{\pi}{2} - \frac{1}{r^3+4r}\left[8 - \left(4r^2+8\right)e^{-\frac{\pi}{2}r}\right] + \frac{1}{r^3+2r}\left[1 - \left(1+2r^2\right)e^{-\pi r}\right] < \frac{\pi G_i}{2G_c^{\mathrm{A}}} \tag{9}$$

with $r = \frac{l}{l_\phi}$.

The curve of $\pi G_i/2G_c^{\mathrm{A}}$ vs. r is shown in Fig. 3b, a smaller ratio of G_i/G_c^{A} requires a smaller value of r. Once values of G_i and G_c^{A} are given, one should choose ratio between l and l_ϕ properly such that point $\left(\log_{10} r, G_i/G_c^{\mathrm{A}}\right)$ fall in the left side area of the curve to meet the requirement in Eq. (9).

With the definition of Eq. (6), Eq. $(4)_1$ can be rewritten as follows

$$G_c(x) = \left[1 - (1-\phi(x))^2\right]\overline{G}_c^i + (1-\phi(x))^2 G_c^{\mathrm{A}} \tag{10}$$

2.3. Kinematics

Restricting the present study to infinitesimal strains, the strain tensor is defined as

$$\boldsymbol{\varepsilon} := \frac{1}{2}(\nabla \boldsymbol{u} + \boldsymbol{u}\nabla) \tag{11}$$

with $\boldsymbol{\varepsilon}: \Omega \to \mathbb{R}^{\dim \times \dim}$, $\nabla(\cdot)$ and $(\cdot)\nabla$ are the left and right gradients, respectively. For a chemo-thermo-mechanical coupling problem, the total strain $\boldsymbol{\varepsilon}$ can be divided into elastic strain $\boldsymbol{\varepsilon}_e$, thermal strain $\boldsymbol{\varepsilon}_\theta$ (resulted from thermal expansion) and growth strain $\boldsymbol{\varepsilon}_n$ (resulted from chemical processes), *i.e.*

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_e + \boldsymbol{\varepsilon}_\theta + \boldsymbol{\varepsilon}_n \tag{12}$$

where

$$\boldsymbol{\varepsilon}_\theta = \alpha_\theta (\theta - \theta_0)\boldsymbol{I} \tag{13}$$

and

$$\boldsymbol{\varepsilon}_n = \alpha_n n \boldsymbol{I} \tag{14}$$

where α_θ is the CTE, $\theta - \theta_0$ is the temperature change from the reference temperature θ_0 to the current temperature θ, $\boldsymbol{I}$ is the second-order identity tensor, α_n is the swelling coefficient of the TGO, and n is the volume ratio of TGO [73]. n measures the extent of oxidation of the BC material, *i.e.* when: (a) $n = 0$, the area is composed completely of BC material; (b) $n \in (0, 1)$, the area is undergoing an oxidation process and is mixed with BC and TGO materials; (c) $n = 1$, the area is completely oxidized and composed solely of TGO material. The time derivative of the total strain is

$$\dot{\boldsymbol{\varepsilon}} = \dot{\boldsymbol{\varepsilon}}_e + \alpha_\theta \dot{\theta} \boldsymbol{I} + \alpha_n \dot{n} \boldsymbol{I} \tag{15}$$

where $(\dot{}) = \frac{\mathrm{d}}{\mathrm{d}t}(\ )$ for brevity.

2.4. Balance equations

2.4.1. Macro-force balance

With the continuum and mechanical boundaries described in Section 2.1. The global balance of linear momentum is specified by

$$\int_\Omega \rho \dot{\boldsymbol{v}} \, \mathrm{d}V = \int_{\partial \Omega_t} \boldsymbol{t} \, \mathrm{d}A + \int_\Omega \boldsymbol{b} \, \mathrm{d}V \tag{16}$$

where ρ is the density, $\boldsymbol{v}$ is the velocity. With Cauchy's stress theorem $\boldsymbol{t} = \boldsymbol{\sigma} \cdot \boldsymbol{n}$, divergence theorem and considering that the volume $\mathrm{d}V$ is arbitrary, leading to the Cauchy's first equation of motion in the local form

$$\rho \dot{\boldsymbol{v}} = \nabla \cdot \boldsymbol{\sigma} + \boldsymbol{b} \tag{17}$$

2.4.2. Micro-force momentum balance

To derive the phase field balance equation, we adopt the micro-force balance law [74–77]. The micro-force momentum balance is given by

$$\int_{\partial \Omega_\chi} \chi \dot{d} \, \mathrm{d}A - \int_\Omega \boldsymbol{\xi} \cdot \nabla \dot{d} \, \mathrm{d}V - \int_\Omega \pi \dot{d} \, \mathrm{d}V = 0 \tag{18}$$

where χ is a scalar micro-traction, π is a scalar micro-stress, $\boldsymbol{\xi}$ is a vector micro-stress. Using divergence theorem, the above equation can be rewritten as

$$\int_{\partial \Omega_\chi} (\chi - \boldsymbol{\xi} \cdot \boldsymbol{n}) \, \dot{d} \, \mathrm{d}A + \int_\Omega (\nabla \cdot \boldsymbol{\xi} - \pi) \cdot \dot{d} \, \mathrm{d}V = 0 \tag{19}$$

Given that $\dot{d}$ is arbitrary, the above equation can be localized as

$$\chi = \boldsymbol{\xi} \cdot \boldsymbol{n} \tag{20}$$

and

$$\nabla \cdot \boldsymbol{\xi} - \pi = 0 \tag{21}$$

2.4.3. Mass balance

For a specific chemical species (oxygen in this study), the number of molecular per unit volume is described by the concentration c, and the number of molecules diffusing across unit area $\mathrm{d}A$ per unit time is described by diffusion flux $\boldsymbol{j}_n$. The mass balance in the presence of chemical reaction can be given as

$$\int_\Omega \dot{c} \, \mathrm{d}V = -\int_{\partial \Omega} \boldsymbol{j}_n \cdot \boldsymbol{n} \, \mathrm{d}A + \int_\Omega r_n \, \mathrm{d}V \tag{22}$$

where r_n is the molecules formation (>0) or consumption (<0) per unit volume per unit time during the chemical reaction. By using divergence theorem and considering the arbitrariness of the volume $\mathrm{d}V$, the local mass balance can be obtained as follows

$$\dot{c} = -\nabla \cdot \boldsymbol{j}_n + r_n \tag{23}$$

2.5. Thermodynamic laws

According to the first law of thermodynamics, the following balance equation of energy is given

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega}\left(w+\frac{1}{2}\rho\dot{\boldsymbol{v}}^2\right)\mathrm{d}V=\int_{\Omega}\boldsymbol{b}\cdot\boldsymbol{v}\,\mathrm{d}V+\int_{\partial\Omega_t}\boldsymbol{t}\cdot\boldsymbol{v}\,\mathrm{d}A+\int_{\Omega}\boldsymbol{\xi}\cdot\nabla\dot{d}\,\mathrm{d}V+\int_{\Omega}\pi\dot{d}\,\mathrm{d}V-\int_{\partial\Omega_q}\boldsymbol{j}_q\cdot\boldsymbol{n}\,\mathrm{d}A+\int_{\Omega}r_q\,\mathrm{d}V-\int_{\partial\Omega_{\mathrm{n}}}\mu_n\boldsymbol{j}_n\cdot\boldsymbol{n}\,\mathrm{d}V \tag{24}$$

where w is the volume density of internal energy, r_q is the body heat source per unit volume and per unit time, $\boldsymbol{j}_q$ is the heat flux per unit area, $\mu_n\boldsymbol{j}_n$ is the energy flux while μ_n being the chemical potential.

Applying the divergence theorem and considering that the volume is arbitrary, the local form of the balance of energy equation is specified

$$\dot{w}=\boldsymbol{\sigma}:\dot{\boldsymbol{\varepsilon}}-\nabla\cdot\boldsymbol{j}_q+\boldsymbol{\xi}\cdot\nabla\dot{d}+\pi\dot{d}+r_q+\mu_n\dot{c}-\mu_n r_n-\boldsymbol{j}_n\cdot\nabla\mu_n \tag{25}$$

The second law of thermodynamics postulates the total production of entropy s for all thermodynamic processes is never negative, then we have Clausius–Duhem inequality in the following global form [78]

$$\frac{\mathrm{D}}{\mathrm{D}t}\int_{\Omega}s\,\mathrm{d}V+\int_{\partial\Omega_q}\frac{\boldsymbol{j}_q}{\theta}\cdot\boldsymbol{n}\,\mathrm{d}A-\int_{\Omega}\frac{r_q}{\theta}\,\mathrm{d}V\geq 0 \tag{26}$$

By virtue of divergence theorem and considering that the integration volume is arbitrary, we have the local form of Clausius–Duhem inequality

$$\theta\dot{s}+\nabla\cdot\boldsymbol{j}_q-\frac{1}{\theta}\boldsymbol{j}_q\cdot\nabla\theta-r_q\geq 0 \tag{27}$$

By elimination the heat source in above equation with Eq. (25), we have the equivalent form

$$\theta\dot{s}-\frac{1}{\theta}\boldsymbol{j}_q\cdot\nabla\theta-\dot{w}+\boldsymbol{\sigma}:\dot{\boldsymbol{\varepsilon}}+\boldsymbol{\xi}\cdot\nabla\dot{d}+\pi\dot{d}+\mu_n\dot{c}-\mu_n r_n-\boldsymbol{j}_n\cdot\nabla\mu_n\geq 0 \tag{28}$$

Introducing the Helmholtz free energy per unit volume

$$\psi=w-\theta s \tag{29}$$

we can rewrite Eq. (28) in the following forms

$$-\frac{1}{\theta}\boldsymbol{j}_q\cdot\nabla\theta-\dot{\psi}-\dot{\theta}s+\boldsymbol{\sigma}:\dot{\boldsymbol{\varepsilon}}+\boldsymbol{\xi}\cdot\nabla\dot{d}+\pi\dot{d}+\mu_n\dot{c}-\mu_n r_n-\boldsymbol{j}_n\cdot\nabla\mu_n\geq 0 \tag{30}$$

The rate of molecules formation r_n is related to the rate of oxidation reaction through the following equation according to [8]

$$r_n=-M\frac{\mathrm{d}n}{\mathrm{d}t} \tag{31}$$

where M is the number of moles of oxygen consumed when TGO is formed per unit volume. We assume that the free energy is a scalar function of the state variables $\{\boldsymbol{\varepsilon}_e,\ \theta,\ d,\ \nabla d,\ c\}$, i.e. $\psi=\psi\left(\boldsymbol{\varepsilon}_e,\ \theta,\ d,\nabla d,\ c\right)$, then

$$\dot{\psi}=\frac{\partial\psi}{\partial\boldsymbol{\varepsilon}_e}:\dot{\boldsymbol{\varepsilon}}_e+\frac{\partial\psi}{\partial\theta}\dot{\theta}+\frac{\partial\psi}{\partial d}\dot{d}+\frac{\partial\psi}{\partial\nabla d}\cdot\nabla\dot{d}+\frac{\partial\psi}{\partial c}\dot{c} \tag{32}$$

Substituting Eqs. (15), (31) and (32) into the dissipation inequality Eq. (30), we have

$$\left(\boldsymbol{\sigma}-\frac{\partial\psi}{\partial\boldsymbol{\varepsilon}_e}\right):\dot{\boldsymbol{\varepsilon}}_e+\left(\boldsymbol{\sigma}:\alpha_\theta\boldsymbol{I}-\frac{\partial\psi}{\partial\theta}-s\right)\dot{\theta}+\left(\boldsymbol{\xi}-\frac{\partial\psi}{\partial\nabla d}\right)\cdot\nabla\dot{d}+\left(\pi-\frac{\partial\psi}{\partial d}\right)\dot{d}+\left(\mu_n-\frac{\partial\psi}{\partial c}\right)\dot{c}+\left(\boldsymbol{\sigma}:\alpha_n\boldsymbol{I}+\mu_n M\right)\dot{n}-\frac{1}{\theta}\boldsymbol{j}_q\cdot\nabla\theta-\boldsymbol{j}_n\cdot\nabla\mu_n\geq 0 \tag{33}$$

Assuming that the dissipation rate is independent of $\left\{\dot{\boldsymbol{\varepsilon}}_e,\ \dot{\theta},\ \nabla\dot{d},\ \dot{c}\right\}$, we have the following constitutive equations and dissipation equations:

$$\begin{cases} \boldsymbol{\sigma} = \dfrac{\partial \psi}{\partial \boldsymbol{\varepsilon}_e} \\ s = \boldsymbol{\sigma} : \alpha_\theta \boldsymbol{I} - \dfrac{\partial \psi}{\partial \theta} \\ \mu_n = \dfrac{\partial \psi}{\partial c} \\ \boldsymbol{\xi} = \dfrac{\partial \psi}{\partial \nabla d} \\ \left(\pi - \dfrac{\partial \psi}{\partial d}\right) \dot{d} \geq 0 \\ \left(\boldsymbol{\sigma} : \alpha_n \boldsymbol{I} + \mu_n M\right) \dot{n} \geq 0 \\ -\dfrac{1}{\theta} \boldsymbol{j}_q \cdot \nabla\theta \geq 0 \\ -\boldsymbol{j}_n \cdot \nabla\mu_n \geq 0 \end{cases} \tag{34}$$

Assuming that the free energy can be decomposed into four parts, elastic energy ψ_e, fracture energy ψ_d, thermal energy ψ_θ, chemical energy related to diffusion ψ_c

$$\psi = \psi_e\left(\boldsymbol{\varepsilon}_e,\ d\right) + \psi_\theta\left(\theta,\ d\right) + \psi_d\left(d, \nabla d\right) + \psi_c\left(c\right) \tag{35}$$

2.5.1. Linear constitutive relations

For hyperelastic materials, the initial elastic energy density can be expressed as

$$\psi_0 = \frac{1}{2} \boldsymbol{\varepsilon}_e : \mathbb{C} : \boldsymbol{\varepsilon}_e \tag{36}$$

where $\mathbb{C} = \frac{\partial \boldsymbol{\sigma}}{\partial \boldsymbol{\varepsilon}_e}$ is the elastic tensor. Considering that the fracture process acts only to the tensile part, the stored energy density is given [79]

$$\psi_e\left(\boldsymbol{\varepsilon}_e,\ d\right) = \omega\left(d\right) \psi_0^+ + \psi_0^- \tag{37}$$

where $\omega\left(d\right)$ is the degradation function. The positive part ψ_0^+ due to tension and negative part ψ_0^- due to compression can be defined as

$$\begin{aligned} \psi_0^+ &= \frac{1}{2}\lambda \left\langle \mathrm{tr}\boldsymbol{\varepsilon}_e \right\rangle_+^2 + \mu \mathrm{tr}\left(\left\langle \boldsymbol{\varepsilon}_e \right\rangle_+^2\right) \\ \psi_0^- &= \frac{1}{2}\lambda \left\langle \mathrm{tr}\boldsymbol{\varepsilon}_e \right\rangle_-^2 + \mu \mathrm{tr}\left(\left\langle \boldsymbol{\varepsilon}_e \right\rangle_-^2\right) \end{aligned} \tag{38}$$

where λ and μ are the Lamé constants. $\langle x \rangle_\pm := \left(x \pm |x|\right)/2$ is the Macaulay bracket, and $\left\langle \boldsymbol{\varepsilon}_e \right\rangle_\pm = \sum_{i=1}^{\dim} \left\langle \varepsilon_i \right\rangle_\pm \boldsymbol{n}_i \otimes \boldsymbol{n}_i$, $\{\varepsilon_i\}_{i=1,\ldots,\dim}$ are the principal strains and $\{\boldsymbol{n}_i\}_{i=1,\ldots,\dim}$ are their eigenvectors.

Applying the degradation to the tensile part of the energy and considering Eq. $(34)_1$, we have

$$\boldsymbol{\sigma} = \omega\left(d\right) \frac{\partial \psi_0^+}{\partial \boldsymbol{\varepsilon}_e} + \frac{\partial \psi_0^-}{\partial \boldsymbol{\varepsilon}_e} \tag{39}$$

By applying the degradation function to the tensile part, this model can prevent unrealistic damage evolving under compression. We adopt the hybrid formulation proposed by Ambati et al. [20] during calculation, which simplifies the calculation for stress without losing accuracy with $\boldsymbol{\sigma} = \omega\left(d\right) \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}_e}$ instead of Eq. (39).

2.5.2. Evolution equation

Assuming that each individual dissipation process decouples with each other (One can also establish a more sophisticated model by considering the cross-coupling effect between different dissipative processes [80]), it can be

derived from the laws of thermodynamics that

$$\begin{cases} -\dfrac{1}{\theta}\boldsymbol{j}_q \cdot \nabla\theta \geq 0 \\ -\boldsymbol{j}_n \cdot \nabla\mu_n \geq 0 \\ (\boldsymbol{\sigma} : \alpha_n \boldsymbol{I} + \mu_n M)\,\dot{n} \geq 0 \\ \left(\pi - \dfrac{\partial\psi}{\partial d}\right)\dot{d} \geq 0 \end{cases} \tag{40}$$

To meet the dissipation requirement of Eq. $(40)_1$, the Fourier's law for the thermal dissipation is adopted

$$\boldsymbol{j}_q = -\boldsymbol{\kappa} \cdot \nabla\theta \tag{41}$$

where $\boldsymbol{\kappa}$ is the thermal conductivity tensor, and will reduce to a scalar thermal conductivity κ for isotropic materials in this paper. Following the treatment similar to [75,81], and considering the degradation on thermal conductivity κ, we have temperature governing equation (More details can be found in Appendix A)

$$\rho C\dot{\theta} = \omega(d)\,\kappa\,\Delta\theta + r_q \tag{42}$$

where C is the specific heat capacity. To meet the dissipation requirement of Eq. $(40)_2$, the Fick's law is adopted for diffusion dissipation

$$\boldsymbol{j}_n = -m\nabla\mu_n \tag{43}$$

where m is a positive parameter. The chemical potential μ_n can be calculated with the following relation [8]

$$\mu_n = \mathcal{N}c \tag{44}$$

where $\mathcal{N}$ is the chemical modulus, inserting Eqs. (31) and (44) into Eq. (23), we have reaction–diffusion equation

$$\dot{c} = D\Delta c - M\frac{\mathrm{d}n}{\mathrm{d}t} \tag{45}$$

where $D = m\mathcal{N}$ is the diffusion coefficient. To meet the dissipation requirement of Eq. $(40)_3$, we can assume the TGO growth rate as

$$\dot{n} = k_0\,(\sigma : \alpha_n \boldsymbol{I} + \mu_n M) \tag{46}$$

where k_0 is a positive parameter, and we use $k_1 = k_0\,(1-n)$ from [8] to limit the growth rate of TGO. Ignoring the strain term, and consider the degradation on k_1, we have

$$\dot{n} = \omega(d)\,G\,(1-n)\,c \tag{47}$$

where $G = k_1\mathcal{N}M$.

With the definition of crack surface density in Eq. (1). The fracture energy density can be given as

$$\psi_d = G_c\gamma \tag{48}$$

The micro-stress $\boldsymbol{\xi}$ and π can be obtained by substituting Eq. (48) into Eq. $(34)_4$ and $(34)_5$ as

$$\boldsymbol{\xi} = \frac{2G_c l}{c_0}\nabla d, \qquad \pi = \omega'(d)\,\psi_0^+ + \frac{\partial\psi_\theta}{\partial d} + \frac{G_c}{c_0 l}\alpha'(d) + \eta\dot{d} \tag{49}$$

where η is a positive parameter. Neglecting the temperature energy term and substituting (49) into Eq. (21) yields the governing equation of phase field

$$Y - \left(G_c\delta_d\gamma + \eta\dot{d}\right) = 0 \tag{50}$$

where $Y = -\omega'(d)\,\psi_0^+$ is the elastic energy driving force, $\delta_d\gamma = \frac{1}{c_0 l}\left(\alpha' - 2l^2\Delta d\right)$ is the functional derivative of crack surface density γ. The above equation can be rewritten as Kuhn–Tucker loading/unloading conditions

$$\dot{d} \geq 0, \quad Y - \left(G_c\delta_d\gamma + \eta\dot{d}\right) \leq 0, \quad \left[Y - \left(G_c\delta_d\gamma + \eta\dot{d}\right)\right]\dot{d} \equiv 0 \tag{51}$$

To prevent the crack from self-healing and guarantee the irreversibility condition ($\dot{d} \geq 0$), we adopt a history maximum energy density $\mathcal{H}$ to replace ψ_0^+ in the energetic driving force Y [82]. The expression of $\mathcal{H}$ is given as

$$\mathcal{H} = \max_{\tau \in [0,t]} \left\{ \psi_0^+, \frac{f_t^2}{2E} \right\} \tag{52}$$

where f_t is the tensile strength and E is the Young's modulus.

Following the work of Wu [52], a specific cohesive laws can be approximated by properly elaborate the following degradation function $\omega(d)$

$$\omega(d) = \frac{(1-d)^p}{(1-d)^p + Q(d)}, \text{ with } Q(d) = a_1 d\left[1 + a_2 d\left(1 + a_3 d\left(\cdots\right)\right)\right] \tag{53}$$

where $p > 0$ is the exponent and a_i are coefficients that determined by the desired traction separation law. We adopt a Cornelissen softening relationship [83] in this work by adopting the degradation function with the following parameters

$$p = 2, \quad a_1 = \frac{4}{\pi}\frac{EG_c}{f_t^2}, \quad a_2 = 1.3868, \quad a_3 = 0.6567 \tag{54}$$

3. Discretization

In this section, the spatial and temporal discretization of the problem is introduced. The whole problem's calculation can be divided into two parts, the first part is the calculation of multi-field coupling problem, which is a transient state problem and both spatial and temporal discretization are needed. The second is the calculation of the interface phase field, which is a steady state problem and only spatial discretization is needed.

3.1. Chemo-thermo-mechanical coupled problem

3.1.1. Spatial discretization

We give here directly the finite element equations for the proposed model, the strong and weak form of the governing equations can be found in Appendix B. The field variables can be discretized as follows

$$\begin{aligned}
&\boldsymbol{u} = \sum_i [N_u]^{\mathrm{T}} \{u\}^i, \ \theta = \sum_i [N_\theta]^{\mathrm{T}} \{\theta\}^i, \ d = \sum_i [N_d]^{\mathrm{T}} \{d\}^i, \ c = \sum_i [N_c]^{\mathrm{T}} \{c\}^i \\
&\boldsymbol{\varepsilon} = \sum_i [B_u]^{\mathrm{T}} \{u\}^i, \nabla\theta = \sum_i [B_\theta]^{\mathrm{T}} \{\theta\}^i, \nabla d = \sum_i [B_d]^{\mathrm{T}} \{d\}^i, \nabla c = \sum_i [B_c]^{\mathrm{T}} \{c\}^i
\end{aligned} \tag{55}$$

where $\{\ \}^i$ denotes the corresponding nodal vectors for element i, $[\ ]^{\mathrm{T}}$ is the transpose of a matrix, $[N_u]$, $[N_\theta]$, $[N_d]$, $[N_c]$ are the shape function matrices for displacement, temperature, phase field and oxygen concentration, respectively, and $[B_u]$, $[B_\theta]$, $[B_d]$, $[B_c]$ are their derivatives.

The governing equations of each field are specified as follows. For displacement field

$$[M]\{\ddot{u}\} = \left\{F_u^{\mathrm{ext}}\right\} - \left\{F_u^{\mathrm{int}}\right\} \tag{56}$$

where

$$[M] = \sum_i \int_{\Omega^i} \rho [N_u]^T [N_u] \, \mathrm{d}V \tag{57}$$

$$\left\{F_u^{\mathrm{int}}\right\} = \sum_i \int_{\Omega^i} \omega(d) [B_u]^T [D] (\{\varepsilon\} - \{\varepsilon_\theta\} - \{\varepsilon_n\}) \, \mathrm{d}V \tag{58}$$

$$\left\{F_u^{\mathrm{ext}}\right\} = \sum_i \left(\int_{\Omega^i} [N_u]^T \{b\} \, \mathrm{d}V + \int_{\partial\Omega_t^i} [N_u]^T \{t\} \, \mathrm{d}A \right) \tag{59}$$

with $[M]$ being the mass matrix, $\left\{F_u^{\mathrm{int}}\right\}$ and $\left\{F_u^{\mathrm{ext}}\right\}$ are the internal force vector and external force vector, respectively, $[D]$ is the material elastic matrix, $\{b\}$ and $\{t\}$ are the nodal body force vector and traction vector, respectively, $\{\varepsilon\}$, $\{\varepsilon_\theta\}$ and $\{\varepsilon_n\}$ are the total strain, thermal strain and growth strain vector, respectively.

For temperature field

$$[C]\left\{\dot{\theta}\right\} = \left\{F_\theta^{\text{ext}}\right\} - \left\{F_\theta^{\text{int}}\right\} \tag{60}$$

where

$$[C] = \sum_i \int_{\Omega^i} \rho C \left[N_\theta\right]^T \left[N_\theta\right] \mathrm{d}V \tag{61}$$

$$\left\{F_\theta^{\text{int}}\right\} = \sum_i \int_{\Omega^i} \omega(d)\, \kappa \left[B_\theta\right]^T \left[B_\theta\right] \{\theta\} \mathrm{d}V \tag{62}$$

$$\left\{F_\theta^{\text{ext}}\right\} = \sum_i \left(\int_{\Omega^i} \left[N_\theta\right]^T \left\{r_q\right\} \mathrm{d}V + \int_{\partial \Omega_q^i} \left[N_u\right]^T \left\{j_q\right\} \mathrm{d}A \right) \tag{63}$$

with $[C]$ being the thermal capacitance matrix, $\left\{F_\theta^{\text{int}}\right\}$ and $\left\{F_\theta^{\text{ext}}\right\}$ are the internal and external thermal power vector, respectively, $\left\{r_q\right\}$ and $\left\{j_q\right\}$ are the heat source intensity vector and external heat flux vector respectively.

For phase field

$$[H]\left\{\dot{d}\right\} = \{Y\} - \{F_d\} \tag{64}$$

where

$$[H] = \sum_i \int_{\Omega^i} \eta \left[N_d\right]^T \left[N_d\right] \mathrm{d}V \tag{65}$$

$$\{Y\} = \sum_i \int_{\Omega^i} \omega'(d) \left[N_d\right]^T \left[N_d\right] \mathcal{H}\, \mathrm{d}V \tag{66}$$

$$\{F_d\} = \sum_i \left(\int_{\Omega^i} \frac{G_c}{c_0 l} \alpha' \left[N_d\right]^T \left[N_d\right] \mathrm{d}V + \int_{\Omega^i} \frac{2 G_c l}{c_0} \left[B_d\right]^{\mathrm{T}} \left[B_d\right] \{d\} \mathrm{d}V \right) \tag{67}$$

with $[H]$ being the artificial viscosity matrix, $\{Y\}$ is the energetic driving force vector, $\{F_d\}$ is the internal fracture energy vector.

For the oxygen concentration field

$$\{\dot{c}\} = \left\{F_c^{\text{diffusion}}\right\} - \left\{F_c^{\text{reaction}}\right\} \tag{68}$$

where

$$\left\{F_c^{\text{diffusion}}\right\} = \left(\sum_i \int_{\Omega^i} D \left[B_c\right]^T \left[B_c\right] \{c\} \mathrm{d}V + \int_{\partial \Omega_n^i} \left[N_c\right]^T \{j_n\} \mathrm{d}A \right) \tag{69}$$

$$\left\{F_c^{\text{reaction}}\right\} = \sum_i \left(- \int_{\Omega^i} \omega(d)\, MG\, (1-n) \left[N_c\right]^{\mathrm{T}} \{c\} \mathrm{d}V \right) \tag{70}$$

with $\left\{F_c^{\text{diffusion}}\right\}$ is the total diffusion flux vector, $\left\{F_c^{\text{reaction}}\right\}$ is the formation or consumption vector due to chemical reaction, $\{j_n\}$ is the external diffusion flux vector.

3.1.2. Temporal discretization

For a time span $[0, T]$, it is divided into $N-1$ time intervals $[t_n, t_{n+1}]$ $(n = 1, 2, \ldots, N-1)$, $t_1 = 0$ is the initial moment and $\Delta t_n = t_{n+1} - t_n$ is the ith interval length. For a time dependent variable x, we use x_{n-1}, x_n and x_{n+1} to denote $x(t_{n-1})$, $x(t_n)$ and $x(t_{n+1})$, respectively. Applying the central difference method to $\{\ddot{u}\}$, we have

$$\{\ddot{u}\}_n = \frac{1}{\Delta t_n^2} \left(\{u\}_{n-1} - 2\{u\}_n + \{u\}_{n+1} \right) \tag{71}$$

and the forward difference method to $\left\{\dot{\theta}\right\}$, $\left\{\dot{d}\right\}$ and $\{\dot{c}\}$, we have

$$\begin{cases} \left\{\dot{\theta}\right\}_n = \dfrac{\{\theta\}_{n+1} - \{\theta\}_n}{\Delta t_n} \\ \left\{\dot{d}\right\}_n = \dfrac{\{d\}_{n+1} - \{d\}_n}{\Delta t_n} \\ \{\dot{c}\}_n = \dfrac{\{c\}_{n+1} - \{c\}_n}{\Delta t_n} \end{cases} \tag{72}$$

Solutions for the moment t_{n+1} can be obtained as follows

$$\{u\}_{n+1} = [M]^{-1}\left[\Delta t_n^2\left(\left\{F_u^{\text{ext}}\right\} - \left\{F_u^{\text{int}}\right\}\right) - [M]\{u\}_{n-1} + 2[M]\{u\}_n\right] \tag{73}$$

$$\{\theta\}_{n+1} = [C]^{-1}\left[\Delta t_n\left(\left\{F_\theta^{\text{ext}}\right\} - \left\{F_\theta^{\text{int}}\right\}\right) + [C]\{\theta\}_n\right] \tag{74}$$

$$\{d\}_{n+1} = [H]^{-1}\left[\Delta t_n\left(\{Y\} - \{F_d\}\right) + [H]\{d\}_n\right] \tag{75}$$

$$\{c\}_{n+1} = [I]^{-1}\left[\Delta t_n\left(\left\{F_c^{\text{diffusion}}\right\} - \left\{F_c^{\text{reaction}}\right\}\right) + [I]\{c\}_n\right] \tag{76}$$

By adopting the lumped form of matrices $[M]$, $[C]$ and $[H]$, Eqs. (73)–(76) can be solved without the need of factorizing a matrix [84], thus accelerates the calculation. The explicit scheme is conditionally stable, the numerical stability limits of displacement field, temperature field, phase field and concentration field are [85,86]

$$\begin{cases} \Delta t_u \approx \dfrac{h_{\min}}{c_d} \quad \text{with} \quad c_d = \sqrt{\dfrac{\lambda + 2\mu}{\rho}} \\ \Delta t_\theta \approx \dfrac{\rho C h_{\min}^2}{2\kappa} \\ \Delta t_d \approx \dfrac{\eta h_{\min}^2}{2 l G_c} \\ \Delta t_c \approx \dfrac{h_{\min}^2}{2D} \end{cases} \tag{77}$$

respectively, where $h_{\min}$ is the smallest distance between two nodes in one element. The time increment adopted in the simulation is

$$\Delta t = \beta \cdot \min\{\Delta t_u,\ \Delta t_\theta,\ \Delta t_d,\ \Delta t_c\} \tag{78}$$

where a parameter $\beta \in (0, 1)$ is used to ensure the satisfaction of the conditions for numerical stability, and $\beta = 0.5$ is adopted in this work.

3.2. Interface phase field problem

The interface phase field variable can be discretized as follows

$$\phi = \sum_i \left[N_\phi\right]^{\mathrm{T}} \{\phi\}^i, \quad \nabla\phi = \sum_i \left[B_\phi\right]^{\mathrm{T}} \{\phi\}^i \tag{79}$$

$\left[N_\phi\right]$ is the shape function matrix for interface phase field, and $\left[B_\phi\right]$ is the derivative of the shape function, $\{\phi\}$ is the interface phase field vector. The Newton–Raphson algorithm is used for the calculation. In a certain incremental step, the iteration scheme can be written as

$$\{\phi\}_{l+1} = \{\phi\}_l - \left[K_\phi\right]_l^{-1}\left\{r_\phi\right\}_l \tag{80}$$

where the subscript l represents the index of the iteration number, $[\ ]^{-1}$ denotes the inverse of a matrix, residual vector $\left\{r_\phi\right\}$ and tangential stiffness matrices $\left[K_\phi\right]$ can be calculated through the following equations

$$\left\{r_\phi\right\} = \sum_i \left(\int_{\Omega^i} \frac{1}{c_0 l_\phi}\alpha' \left[N_\phi\right]^T \left[N_\phi\right] \mathrm{d}V + \int_{\Omega^i} \frac{2 l_\phi}{c_0} \left[B_\phi\right]^{\mathrm{T}} \left[B_\phi\right]\{\phi\}\ \mathrm{d}V\right) \tag{81}$$

$$\left[K_\phi\right] = \sum_i \left(\int_{\Omega^i} \frac{1}{c_0 l_\phi}\alpha'' \left[N_\phi\right]^T \left[N_\phi\right] \mathrm{d}V + \int_{\Omega^i} \frac{2 l_\phi}{c_0} \left[B_\phi\right]^{\mathrm{T}} \left[B_\phi\right] \mathrm{d}V\right) \tag{82}$$

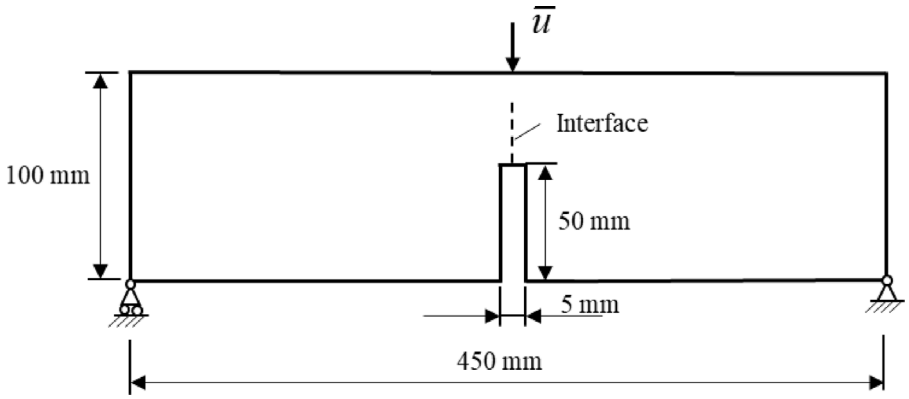


Fig. 4. Three-point bending of a notched concrete beam, boundary conditions and loading (surface represented by dotted line).

In particular, the present work neglects the updating of the interface's position and property changes, which thus has made the problem be linear, the equation can be solved by one time and the iteration can be escaped.

4. Preliminary study

In this section, two parametric studies are conducted to discuss how mesh size h and interface half-bandwidth l_ϕ affect the results, respectively. A three-point bending (TPB) problem is investigated. The geometric setting and boundary conditions are shown in Fig. 4 with an interface on top of the notch. The fracture toughness of the bulk material is 10 times of that of the interface: $G_c^b = 10G_c^i$. The internal length scale of phase field is fixed at $l = 2.5$ mm.

The load–reflection curves and fracture energy curves with different mesh size ($h = 0.1$, 0.2, 0.5, 1.0 mm) are shown in Fig. 5a and b, respectively. The results show that the convergence of load–reflection curves and fracture energy curves are both achieved as the mesh size decreases. Then five cases are conducted to investigate the influence of $r = l/l_\phi$ on the results with a fixed mesh size of $h = 0.2$ mm. The load–reflection curves and fracture energy curves are shown in Fig. 5c and d, respectively. When the dissipation equivalence is considered (G_c is calculated using Eq. (10)), the results change slightly with different values of r (solid lines) and are deviated from the result without considering dissipation equivalence (G_c is calculated using Eq. $(4)_1$). The results demonstrated that the compensation treatment can keep the dissipation equivalence of the interface fracture.

5. Failure process simulation

All the numerical simulations (except for Sections 5.1 and 5.2) use the same material properties listed in Tables 1 and 2. It is noteworthy that, material properties of BC layer will gradually change as the oxidation process goes, thus a specific material property P of BC layer is dependent on the volume ratio n through the following equation

$$P_{\text{eq}} = (1 - n)\, P_{\text{BC}} + n\, P_{\text{TGO}} \tag{83}$$

where P_{eq} is the material property of BC layer during oxidation process, P_{BC} is the initial material property of BC layer when the oxidation has not started yet, P_{TGO} is the material property when the oxidation process has finished. The material properties concerning diffusion and oxidation process are listed in Table 3. Oxygen diffusion and material oxidation of bond coating material are considered in both Sections 5.4 and 5.5. Diffusion and oxidation are calculated only when the temperature is above 850 °C. The temperature is uniformly distributed in Sections 5.3–5.5.

5.1. L-shaped panel test

A L-shaped panel fracture test conducted by Winkler [87] is studied. The geometric setting and boundary conditions are shown in Fig. 6. The material properties are taken from [87] and are shown in Table 4. A finer mesh is assigned to the critical zone, the internal length scale is set to $l = 2.5$ mm.

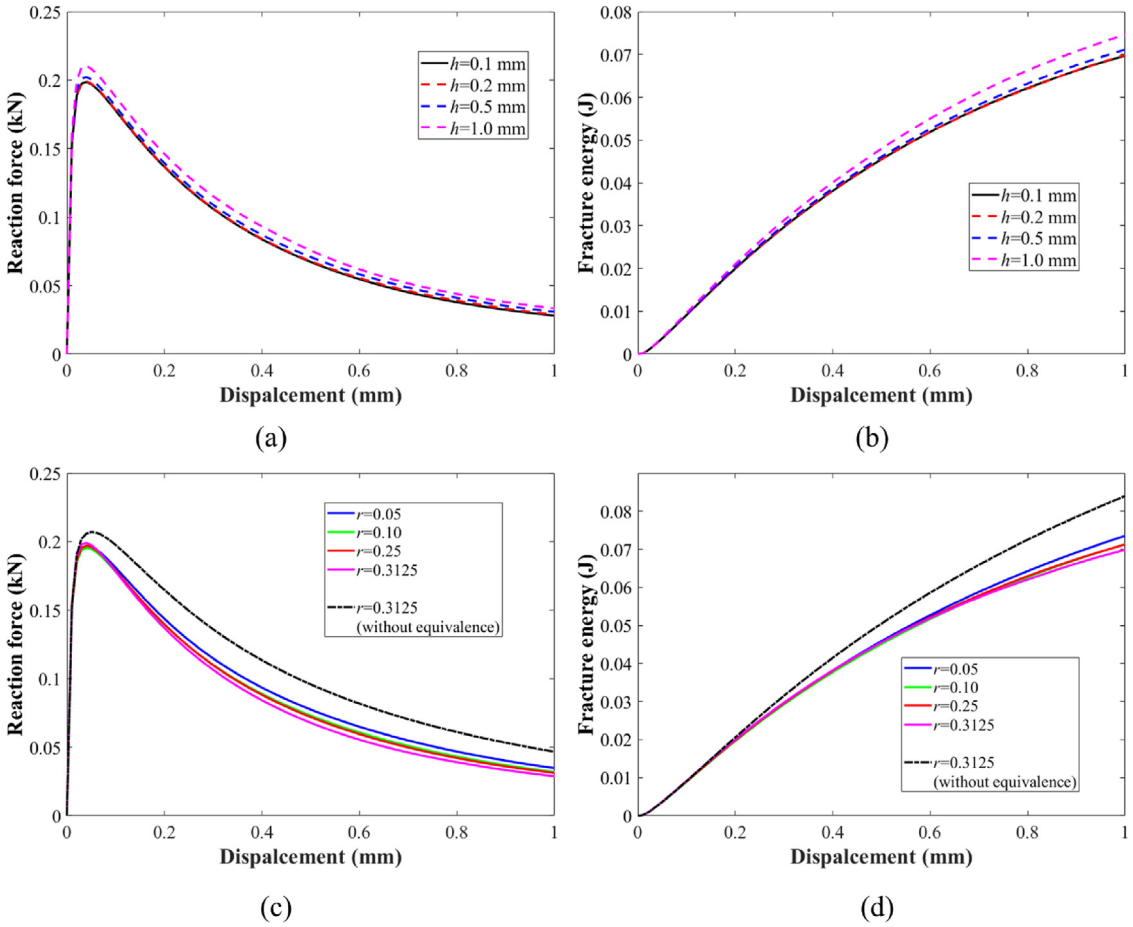


Fig. 5. (a) Load reflection curves and (b) energy dissipation curves with different mesh sizes h. (c) Load reflection curves and (d) energy dissipation curves with different value of r (black dotted lines represent results without considering dissipation equivalence).

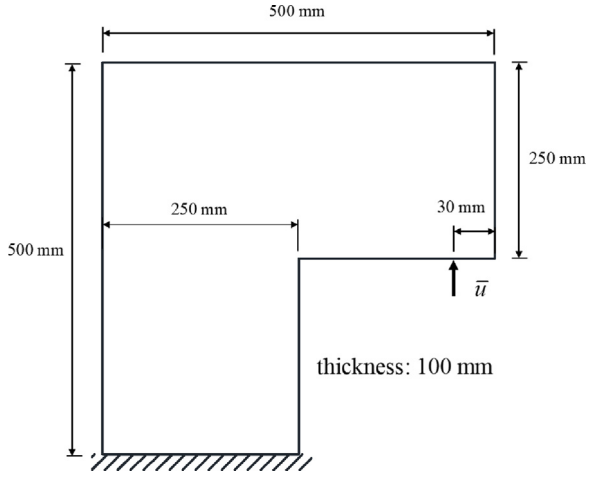


Fig. 6. Geometric setting and boundary conditions of a L-shaped panel, the bottom edge of the specimen is fixed in both x and y directions, a prescribed displacement loading is applied upward at the bottom edge of the right side of the specimen.

Table 1
Temperature-dependent material properties for a TBC system [6].

T (°C)	α $(\times 10^{-6}\ ^{\circ}\mathrm{C}^{-1})$	E (GPa)	ν
Top coat layer			
20	9.68	17.5	0.2
220	9.67	16.3	0.2
420	9.71	15.2	0.2
620	9.81	14.0	0.2
820	10.02	12.9	0.2
1020	10.04	11.7	0.2
Bond coat layer			
20	12.36	152	0.32
220	13.04	151	0.33
420	13.91	145	0.33
620	14.97	132	0.34
820	16.22	109	0.35
1020	17.65	71.9	0.35
Thermally grown oxide			
20	5.08	380	0.27
220	5.90	369	0.27
420	6.73	361	0.27
620	7.55	352	0.27
820	8.38	336	0.27
1020	9.20	309	0.27
Substrate			
20	14	200	0.3
700	15	150	0.3
1000	16.6	150	0.3

Table 2
Temperature-independent material properties.

	ρ $(\mathrm{kg/mm^3})$ [10]	G_c (kN/mm) [15]	f_t (GPa) [15]
Top coat layer	5.65×10^{-6}	8×10^{-6}	0.2
Bond coat layer	7.32×10^{-6}	3×10^{-4}	0.5
Thermally grown oxide	3.978×10^{-6}	6×10^{-5}	0.38
Substrate	8.22×10^{-6}	\	\
TC/TGO interface	\	1×10^{-5}	0.05
TGO/BC interface	\	2×10^{-5}	0.2

Table 3
properties concerning diffusion and oxidation process [73].

Properties	Values
D^{BC} $(\mathrm{m^2\ s^{-1}})$	2.0×10^{-13}
D^{TGO} $(\mathrm{m^2\ s^{-1}})$	2.0×10^{-13}
G $(\mathrm{m^3\ mol^{-1}\ s^{-1}})$	1×10^{-4}
M $(\mathrm{mol\ m^{-3}})$	1.11×10^{5}
α_n	8×10^{-3}

The predicted crack path as well as load–reflection curves are depicted in Fig. 7. The curved crack path predicted using the proposed model agrees well with the experimental observation as shown in Fig. 7a. The peak load and softening behavior of this work match the experimental results well, and also in good agreement with Wu's result [52] which is obtained using a phase field model.

Table 4
Material properties of the L-shaped panel test.

$\rho\,(\mathrm{kg/mm^3})$	E (GPa)	ν	$G_c\ (\mathrm{J/mm^2})$	f_t (GPa)
2×10^{-9}	25.28	0.18	0.09	2.7×10^{-3}

Table 5
Material properties for quenching test.

$\rho\,(\mathrm{kg/m^3})$	E (GPa)	ν	κ (W/ (mm K))	$G_c\ (\mathrm{J/mm^2})$	C (J/(kg K))	α_θ
2.45×10^{-6}	340	0.22	0.3	42.47×10^{-6}	775	8×10^{-6}

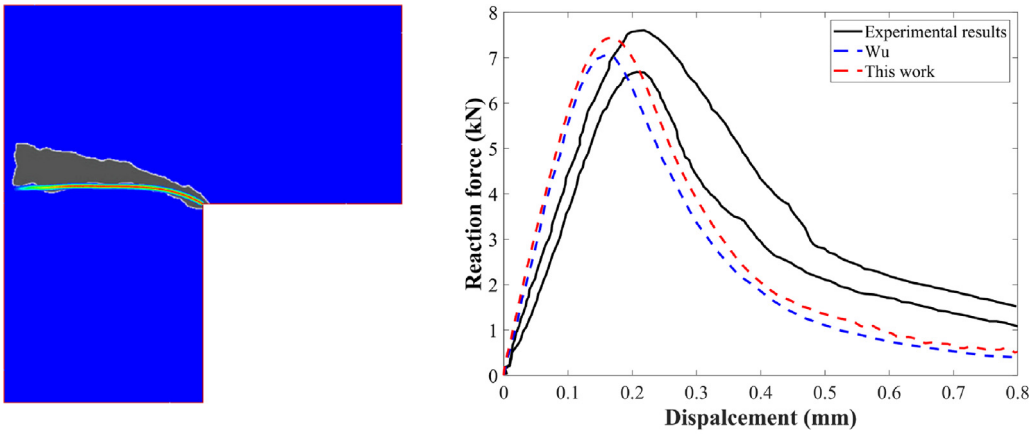


Fig. 7. (a) Fracture pattern predicted by the proposed model and experimental observation (gray area). (b) Load–reflection curves.

5.2. A quenching test

To validate our proposed model, we studied in this section with a quenching test. The calculation of the oxygen concentration field and chemical reaction is not considered in this section. Consider a ceramic plate heated to a high temperature and then put into cold water, leading to the shrinkage of the surface, according to [88], a large number of parallel cracks will occur in the surface of the ceramic.

According to the experimental setting in [88], we take half of the ceramic specimen for simulation, the dimension of the specimen is 50×9.8 mm. The initial temperature of the plate is 680 K, and the surface of the specimen is kept at an ambient temperature of 300 K. The material properties are listed in Table 5.

The mesh has been refined near the interface of the specimen to gain a better accuracy, the refined mesh size is $h=0.02$ mm, the internal length scale is $l=0.08$ mm, the artificial viscosity parameter of phase field is chosen to be $\eta=1\times10^{-8}$ kN s/mm^2. The total period of the physical process is 0.01 s. The simulation results are shown in Fig. 8. As the low temperature imposed on the surface of the specimen, a steep temperature gradient is generated, which leading to a large residual stress and thus multiple parallel cracks formed.

We offered also the experimental observation from [88] for the comparison with our simulation results, the crack density of our results are in good agreement with that of the experimental results, but the length of the long cracks in our results are slightly short than that of the experimental observation, this is because the total simulation period is shorter than the experimental setting of 10 min (see Fig. 9).

5.3. BC/TGO interface delamination

Consider a TBC system with a fixed TGO thickness ($n=1$), and the TGO growth and swelling effect are not considered in this case. The geometric setting and boundary conditions of TBC system in this case is shown

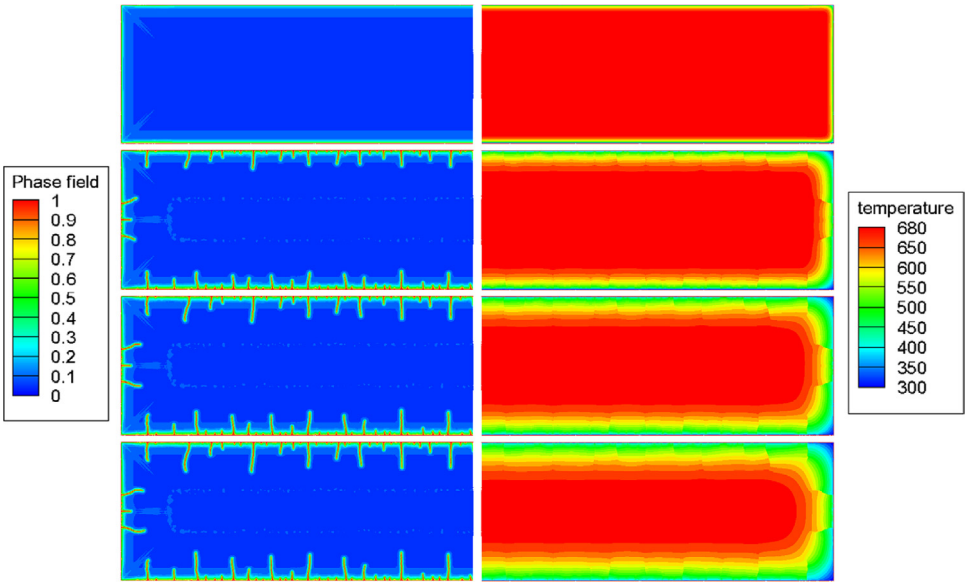


Fig. 8. The simulation results of phase field (left) and temperature (right) distribution at different moments, (a) $t = 0.0025$ s, (b) $t = 0.005$ s, (c) $t = 0.0075$ s, (d) $t = 0.01$ s.

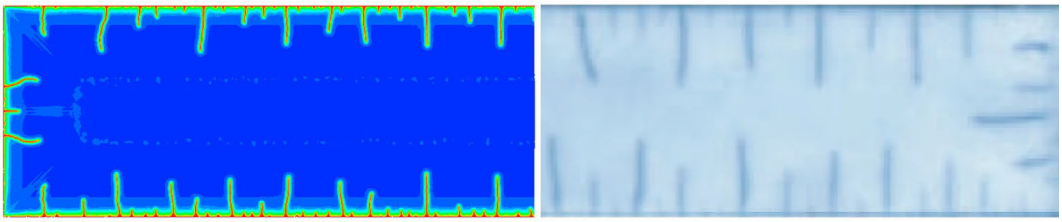


Fig. 9. The simulated fracture pattern (left) at $t = 0.01$ s and experimental observation (right) [88], with a temperature drop of 380 K.

in Fig. 10a, the thickness of TC layer and BC layer are 500 μm and 100 μm, respectively. The thickness of the substrate is 3 mm. The interface between TC and BC is assumed an idealized sinusoidal-shaped curve to model the roughness of the APS TBC interface's topology. The vertical position y of the sinusoidal interface, as a function of the horizontal position x, is expressed as $y(x) = A\sin(2\pi/L)$, where $A = 20$ μm is the amplitude and $L = 60$ μm the wave length (the influence of amplitude of the imperfection on the stress distribution and failure mechanism is well-studied by now and will not discussed further in this work, see literature [6,16,89,90]). We use half-period model for simulation, the translational DOF of the x-direction of the left edge is fixed, and the multipoint constraints are applied to the right edge to ensure that the right edge keeping straight during deformation. The thickness of TGO layer is 5 μm. Interface phase field is used to smear the interface (*i.e.* fracture toughness and strength), the length scale of the interface phase field is $l_\phi = 0.002$. The distribution of the interface phase field is shown in Fig. 10b.

A finer mesh density is assigned to the critical zone near the interface with a mesh size of $h = 0.0002$ mm, and the internal length scale of phase field is $l = 0.0005$ mm. The temperature is uniformly distributed in the TBC system, and cooling down from an elevated temperature of 1150 °C to a room temperature of 20 °C in 0.3 h.

We first investigate the distribution of stress at the end of a cooling process without considering fracture process and damage evolution. The maximum principal residual stress distribution is shown in Fig. 11. During cooling process, tensile stress appears at the off-peak region of the TC layer, while compression stress at the valley of the TC layer (Fig. 11b). Crack may initiate at the off-peak region of the TC layer, and propagate horizontally. There is also significant tensile stress near the interface around the peak region of BC layer and TGO layer (Fig. 11a, c and

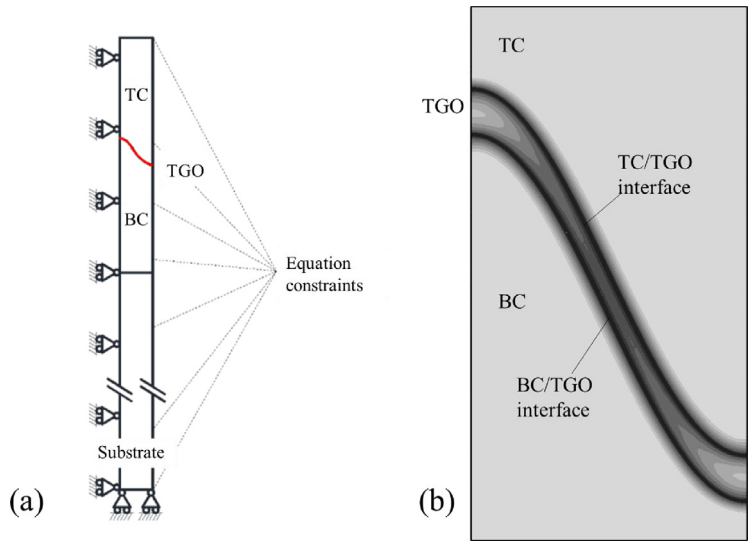


Fig. 10. (a) Geometric setting for a half-period model and its boundary conditions, the equation constraints used to keep the right edge straight during calculation. (b) The distribution of interface phase field.

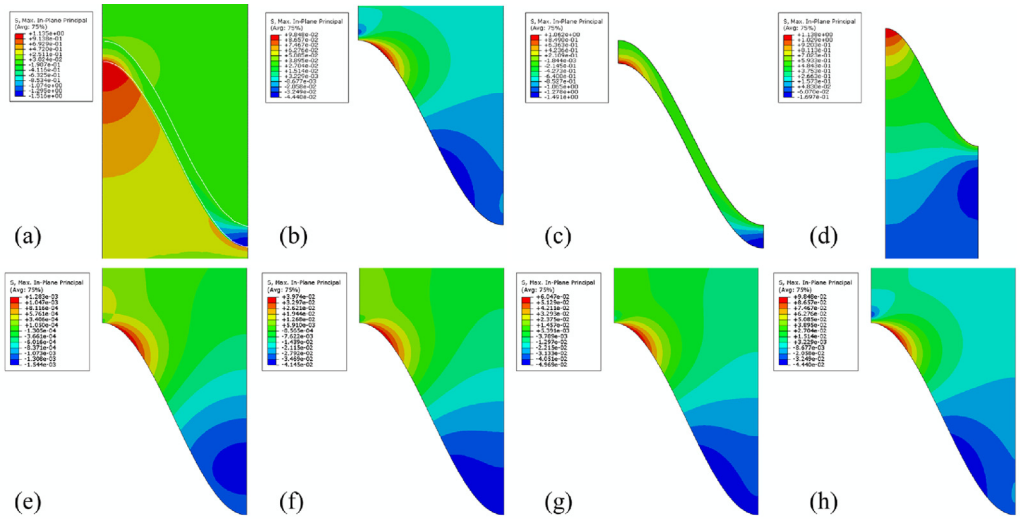


Fig. 11. Stress distribution (y-direction) of (a) a TBC system (white lines denote the interface), (b) TC layer, (c) TGO layer and (d) BC layer at the end of a cooling process, and the distribution of the maximum principal stress of TC layer at time (e) $t = 0.003$ h, (f) $t = 0.150$ h, (f) $t = 0.210$ h, (f) $t = 0.300$ h.

d), which is the key factor causing the interface delamination of the BC/TGO interface. The evolution of maximum principal stress of TC layer is given in Fig. 11e–h.

Then a simulation is carried out on the fracture process under the given loading condition. As can be seen from Fig. 12a–d, as the cooling process proceeds, interfacial delamination occurs at the peak of BC/TGO interface, experimental observation from [91] is offered for validation. The interfacial delamination of BC/TGO interface during cooling process [5,14,92] has been also reported by other authors. The distribution of the maximum principal stress is also presented, by comparing the stress results with and without considering damage, we find that the damage accumulated at the peak of BC/TGO interface can affect the distribution of the stress. There is little difference between the two situations (with and without modeling damage) at the beginning of cooling period (Fig. 12f and g). As the damage is accumulated at the interface, the stress level keeps raising, while the position

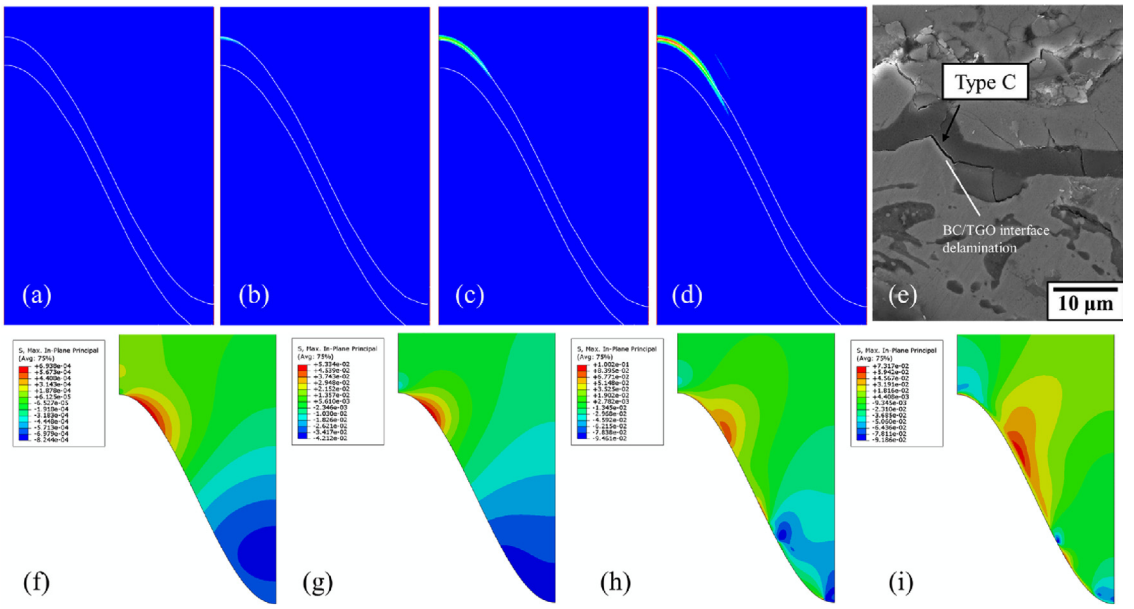


Fig. 12. (a)–(d) Damage initiation and crack propagation process (white lines denote the interface), (e) experimental observation [8], and the distribution of the maximum principal stress of TC layer at time (f) $t = 0.003$ h, (g) $t = 0.150$ h, (h) $t = 0.210$ h, (i) $t = 0.300$ h.

of the maximum value of stress moves along the interface (Fig. 12h). When the interface is delaminated at the peak, the stress distribution is quite different from that in Fig. 11h (both are at the end at the cooling period), the maximum value of the stress moves to the crack tip position, and peak position is relaxed due to crack formation, the compression area at the valley of the interface gradually vanishes due to the crack propagation, and the maximum value of stress is also relaxed.

5.4. Top coating crack

We take a one-period model for simulation, the wave length of the interface roughness is $L = 30$ μm, the amplitude is $A = 5$ μm, and the boundary conditions are the same with that in Fig. 10a. The initial thickness of the TGO layer is 0 μm. The temperature drops from the initial value of 1150 °C to 20 °C in 30 min. Following the work of Zhou et al. [8], we chose the oxygen diffusion coefficients of TBC and BC to be equal for simplicity. We do not consider the oxygen diffusion in TC layer, and a prescribed oxygen concentration is given on TC/TGO interface with a value of 1.55 mol m^{-3} [73]. A finer mesh density is assigned to the critical zone near the interface with a mesh size of $h = 0.0002$ mm, and the internal length scale of phase field is $l = 0.0005$ mm.

The simulation results and experimental observation from [90] are shown in Fig. 13. The cracks are initiated at the peak of the TC/TGO interface due to the mismatch of CTEs, and then propagate along the interface for a short distance, then propagate horizontally at the off-peak position of the interface, until cracks coalesce with each other at the TC layer, and leading to the ultimate spallation failure of the TBC. Furthermore, the horizontally propagated cracks may interact with pre-existing defects (the micro-cracks, pores and voids), facilitates the spallation of the coating. The results are quite different with that of Section 5.3, which can be explained by the difference of TGO thickness. The thickness of TGO layer in Section 5.3 is fixed at 5 μm, while the thickness of the current simulation is nearly 0 μm (we define TGO material by volume ratio $n > 0.8$ in this paper), which clarifies that the interfacial delamination of BC/TGO interface is caused by the mismatch of CTEs between TGO and BC material.

5.5. TC/TGO interface delamination

In this simulation, the TBC system first undergoes an 800 h isothermal period with a temperature of 1150 °C, during which the BC material gradually oxidated and turned into TGO material. And then a cooling period is

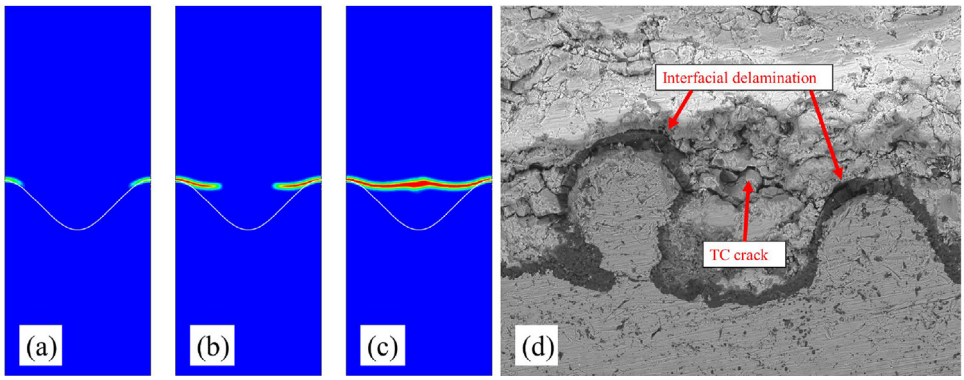


Fig. 13. (a)–(c) Crack initiation, propagation and coalescence, (d) experimental observation [90].

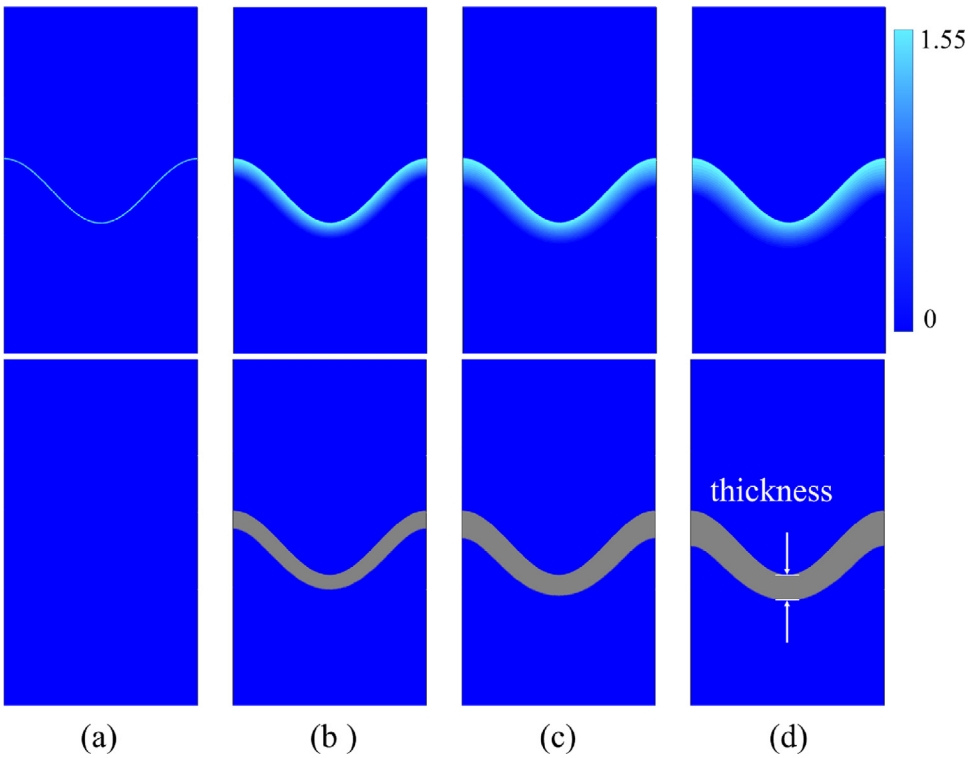


Fig. 14. Oxygen distribution and TGO layer at (a) 0 h, (b) 267 h, (c) 533 h and (d) 800 h. The TGO material is defined by volume ratio $n > 0.8$.

followed, during which the temperature drops from 1150 °C to 20 °C in 30 min. The geometric setting and boundary conditions is identical to that in Section 5.4. The concentration of oxygen and oxidation process are shown in Fig. 14. And the thickness of TGO layer as a function of time is shown in Fig. 15, result of Zhou et al. are provided for comparison [8].

The fracture process is shown in Fig. 16. Damages first initiate from the TC/TGO interface (A, B, C and D points in Fig. 16a) as the oxidation process proceeds during the isothermal period, which is mainly caused by the swelling effect due to oxidation process. During the cooling period, the cracks are nucleated in the previously damaged points (Fig. 16b), the residual stress continue to increase as the temperature decreases, the cracks are gradually propagated (Fig. 16c) and finally formed an interfacial crack (Fig. 16d). It is noteworthy that there is damage initiation at the peak area of the BC/TGO interface during the period, and the cracks of the TC/TGO interface and the damage of

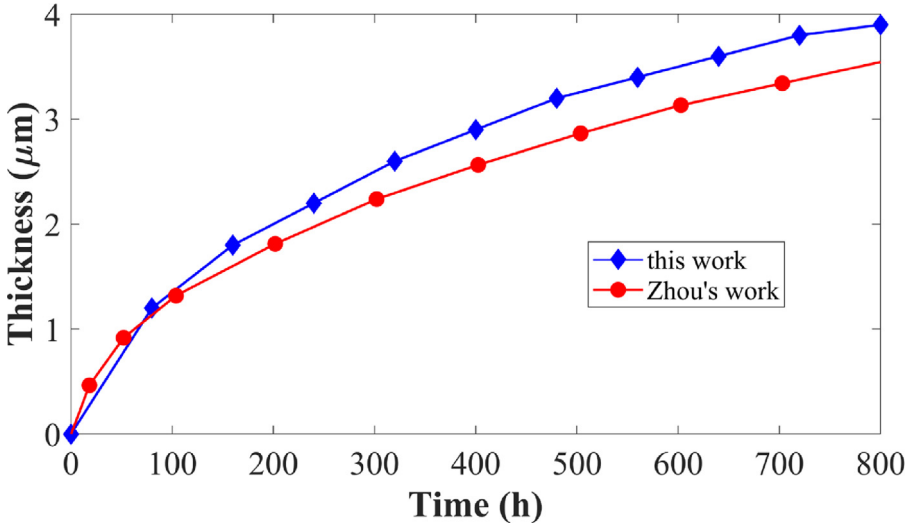


Fig. 15. Thickness of the TGO layer versus time (measured at the valley of the TGO layer as shown in Fig. 14d), results of this work and Zhou [8].

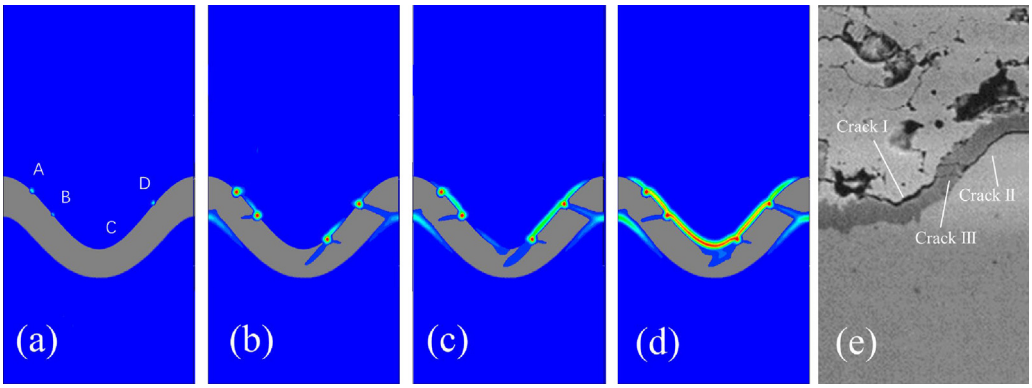


Fig. 16. (a) damage initiation at the end of the isothermal period, (b)–(d) crack nucleation and propagation during the cooling period, (e) experimental observation [91], crack I is the TC/TGO interfacial crack, crack II is the BC/TGO interfacial crack, crack III is the bulk crack inside TGO layer.

the BC/TGO interface are interact with each other and may develop into a bulk crack inside TGO. The simulation results are in good agreement with the experimental observation from [91] (crack III in Fig. 16e).

Fig. 17 shows the changes in maximum principal stress at the peak of the roughness and fracture energy of the whole TBC system during isothermal and cooling period. The maximum principal stress is gradually increasing during the isothermal period, which is caused by the swelling effect of oxidation process, and the fracture energy of the whole domain is kept at a low level because there is little damage caused during the isothermal period. During the cooling period, the stress continues to increase due to the mismatch of CTEs, in the meanwhile, the fracture energy increases because the crack initiation and propagation occurred. At the end of the cooling period, the stress is relaxed due to the damage and delamination and thus prohibit the TC cracks.

6. Conclusion

In this work, a chemo-thermo-mechanical coupled phase field framework is proposed. Interfacial properties are smeared for the calculation of the interfacial delamination, several numerical simulations are conducted and experimental observations are offered for comparison. The following conclusion can be obtained:

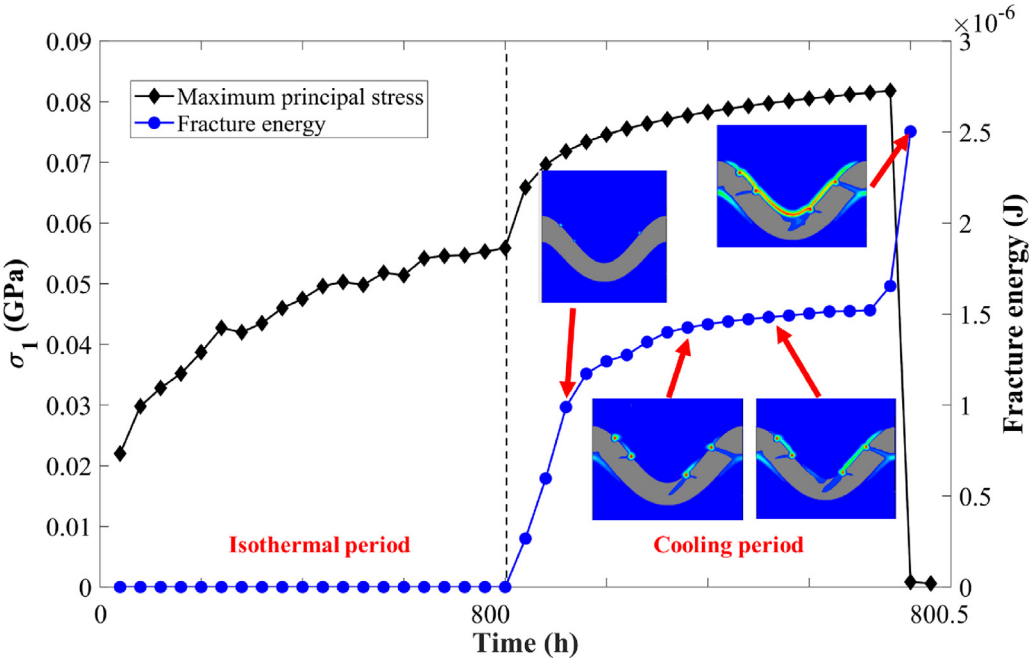


Fig. 17. Maximum principal stress curve time and fracture energy curve, and the corresponding damage and fracture patterns.

● The proposed model can simulate both interfacial and bulk cracking of the TBC system, the BC/TGO interface delamination are observed in the cooling process when the TGO thickness is non-zero. When the TGO thickness is zero or very small, the interface crack appears at the peak of TC/TGO interface and propagated along the interface for a short distance, then transformed into the horizontally propagated crack inside TC layer and coalesces with another crack.

● The swelling of the TGO layer causes the damage initiation at the TC/TGO interface, and formed into cracks at the cooling period, eventually leading to the delamination of the TC/TBC interface. The cracks at the TC/TGO interface and the cracks formed at the BC/TGO interface are interact with each other and may develop further into bulk cracks of the TGO layer.

● The dissipation equivalence of the interface fracture is guaranteed by adopting a modified fracture toughness and validated though a numerical simulation. To prevent the non-physical negative value of the modified fracture toughness, the choice of interface phase field half-bandwidth has been discussed.

The proposed model is a powerful and effective tool for TBC failure simulation and other multi-field problems calculation. Still, there is much room for this model to improve, current model can predict only the cracks caused by tensile stress, different failure criteria can be considered in this model to improve the performance. Besides, the micro-defects in TBC system can also be considered in the future work.

Nomenclature

Variables

$\boldsymbol{u}, \overline{\boldsymbol{u}}, \boldsymbol{v}$	Displacement, prescribed displacement and velocity
$\theta, \overline{\theta}, \theta_0$	Temperature, prescribed displacement and reference temperature
$c, \overline{c}$	Oxygen concentration and prescribed oxygen concentration
$\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon}_e$, $\boldsymbol{\varepsilon}_\theta$, $\boldsymbol{\varepsilon}_n$	Total strain, elastic strain, thermal strain, growth strain tensors
n	Volume ratio
α_θ, α_n	Coefficient of thermal expansion, coefficient of TGO swelling
$\boldsymbol{t}$, $\boldsymbol{b}$	Surface traction and body force vector
$\chi, \pi, \boldsymbol{\xi}$	Scalar micro-traction, scalar and vector micro stress
$d, \nabla d$	Phase field and its gradient

$\Gamma_c, \Gamma_l, \gamma$	Discrete and regularized crack surface functional and crack surface density functional
$\omega(d)$	Degradation function
$\alpha(d)$	Crack geometric function
l, b	Internal length scale parameter, half-bandwidth of the regularization area
$\boldsymbol{j}_n, \boldsymbol{j}_q$	Diffusion and heat flux
r_n, r_q	Molecules formation (>0) or consumption (<0) and body heat source density
w	Volume density of internal energy
s	Entropy
ψ	Helmholtz free energy per unit volume
$\mathbb{C}$	Fourth-order elastic tensor
Material properties	
E	Young's modulus
ν	Poisson's ratio
f_t	Tensile strength
ρ	Density
C	Specific heat capacity
M	Number of moles of oxygen consumed when per unit volume of TGO is formed
D	Diffusion coefficient
η	Artificial viscosity parameter of phase field
κ	Thermal conductivity

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

By virtue of Eq. (29), and (35), the internal energy can be expressed as

$$w = \psi_e + \psi_d + \psi_c + \psi_\theta + \theta s \tag{84}$$

The material time derivative of w is specified by

$$\dot{w} = \dot{\psi}_e + \dot{\psi}_d + \dot{\psi}_c + \dot{\psi}_\theta + \dot{\theta} s + \theta \dot{s} \tag{85}$$

Using Eq. (34)$_2$ and $\dot{\psi}_\theta = \frac{\partial \psi_\theta}{\partial \theta}\dot{\theta} + \frac{\partial \psi_\theta}{\partial d}\dot{d}$, we have

$$\dot{w} = \dot{\psi}_e + \dot{\psi}_d + \dot{\psi}_c + \frac{\partial \psi_\theta}{\partial d}\dot{d} + \boldsymbol{\sigma} : \alpha_\theta \dot{\theta} \boldsymbol{I} + \dot{\boldsymbol{\sigma}} : \boldsymbol{\varepsilon}_\theta + \rho C \dot{\theta} \tag{86}$$

where $\rho C = -\frac{\partial^2 \psi_\theta}{\partial \theta^2}\theta$ with ρ being the density and C the specific heat capacity. Considering the following relations

$$\begin{aligned}
\dot{\psi}_e &= \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}}_e + \frac{\partial \psi_e}{\partial d}\dot{d} \\
\dot{\psi}_d &= \boldsymbol{\xi} : \nabla \dot{d} + \pi \dot{d} \\
\dot{\psi}_c &= \mu_n \dot{c} \\
\dot{\psi}_\theta &= \frac{\partial \psi_\theta}{\partial \theta}\dot{\theta} + \frac{\partial \psi_\theta}{\partial d}\dot{d}
\end{aligned} \tag{87}$$

and Eqs. (25) and (31), yields

$$\rho C\dot{\theta} = -\nabla \cdot \boldsymbol{j}_q + r_q - \dot{\boldsymbol{\sigma}} : \boldsymbol{\varepsilon}_\theta + D_{dis} \tag{88}$$

where the dissipation term

$$D_{dis} = -\boldsymbol{j}_n \cdot \nabla \mu_n + (\boldsymbol{\sigma} : \alpha_n \boldsymbol{I} + \mu_n M)\,\dot{n} - \left(\frac{\partial \psi_e}{\partial d} + \frac{\partial \psi_\theta}{\partial d}\right)\dot{d} \tag{89}$$

characterizing the temperature change due to damage evolution and oxidation.

Substituting Eq. (41) in Eq. (88) and neglecting the dissipation term D_{dis} and $\dot{\boldsymbol{\sigma}} : \boldsymbol{\varepsilon}_\theta$, we have the evolution equation of temperature as

$$\rho C\dot{\theta} = \omega(d)\,\kappa\,\Delta\theta + r_q \tag{90}$$

Appendix B

1.*Strong form of the governing equations*

The strong form of governing equations are summarized as follows

$$\begin{cases} \upsilon_u = \rho\dot{\boldsymbol{v}} - \omega(d)\,\nabla \cdot \boldsymbol{\sigma} - \boldsymbol{b} = 0 \\ \upsilon_\theta = \rho C\dot{\theta} - \omega(d)\,\kappa\,\Delta\theta - r_q = 0 \\ \upsilon_d = Y - \left(G_c\delta_d\gamma + \eta\dot{d}\right) = 0 \\ \upsilon_d = \dot{c} - D\Delta c + M\dfrac{\mathrm{d}n}{\mathrm{d}t} = 0 \end{cases} \quad \text{in } \Omega \tag{91}$$

and the boundaries corresponding to Dirichlet and Neumann type boundary conditions, we have

$$\begin{aligned} &\partial\Omega_u \cup \partial\Omega_t = \partial\Omega, \quad \partial\Omega_u \cap \partial\Omega_t = \varnothing \\ &\partial\Omega_\theta \cup \partial\Omega_q = \partial\Omega, \quad \partial\Omega_\theta \cap \partial\Omega_q = \varnothing \\ &\partial\Omega_c \cup \partial\Omega_n = \partial\Omega, \quad \partial\Omega_c \cap \partial\Omega_n = \varnothing \end{aligned} \tag{92}$$

and the corresponding boundary conditions are

$$\begin{aligned} &\begin{cases} \boldsymbol{u} = \overline{\boldsymbol{u}} & \text{on } \partial\Omega_u \\ \overline{\upsilon}_u = \omega(d)\,\boldsymbol{\sigma} \cdot \boldsymbol{n} - \boldsymbol{t} = 0 & \text{on } \partial\Omega_t \end{cases} \\ &\begin{cases} \theta = \overline{\theta} & \text{on } \partial\Omega_\theta \\ \overline{\upsilon}_\theta = \boldsymbol{j}_q \cdot \boldsymbol{n} - q_n = 0 & \text{on } \partial\Omega_q \end{cases} \\ &\overline{\upsilon}_d = \frac{2G_c l}{c_0}(\nabla d \cdot \boldsymbol{n} - 0) = 0 \text{on } \partial\Omega \\ &\begin{cases} c = \overline{c} & \text{on } \partial\Omega_c \\ \overline{\upsilon}_c = \boldsymbol{j}_n \cdot \boldsymbol{n} - J_n = 0 & \text{on } \partial\Omega_n \end{cases} \end{aligned} \tag{93}$$

2.*Weak form of the governing equations*

The weak form is constructed by multiplying each equation by an appropriate arbitrary test function, integrating over the domain of the problem, as follows

$$R_u = (\upsilon_u,\ \delta\boldsymbol{u})_\Omega + (\overline{\upsilon}_u,\ \delta\boldsymbol{u})_{\partial\Omega_t} = 0 \tag{94}$$

$$R_\theta = (\upsilon_\theta,\ \delta\theta)_\Omega + (\overline{\upsilon}_\theta,\ \delta\theta)_{\partial\Omega_q} = 0 \tag{95}$$

$$R_d = (\upsilon_d,\ \delta d)_\Omega + (\overline{\upsilon}_d,\ \delta d)_{\partial\Omega} = 0 \tag{96}$$

$$R_c = (\upsilon_c,\ \delta c)_\Omega + (\overline{\upsilon}_c,\ \delta c)_{\partial\Omega_n} = 0 \tag{97}$$

where $(\cdot,\cdot)$ denotes the inner product in L^2 space, the following equations can be obtained from Eqs. (94)–(97)

$$\int_\Omega \rho\dot{\boldsymbol{v}} \cdot \delta\boldsymbol{u}\,\mathrm{d}V - \int_\Omega \boldsymbol{b} \cdot \delta\boldsymbol{u}\,\mathrm{d}V + \int_\Omega \omega(d)\,\boldsymbol{\sigma} : \delta\boldsymbol{\varepsilon}\,\mathrm{d}V + \int_{\partial\Omega_t} \boldsymbol{t} \cdot \delta\boldsymbol{u}\,\mathrm{d}A = 0 \tag{98}$$

$$\int_{\Omega} \rho C \dot{\theta} \delta\theta \, \mathrm{d}V + \int_{\Omega} \omega(d) \, \kappa \nabla\theta \cdot \delta\nabla\theta \, \mathrm{d}V - \int_{\Omega} r_q \delta\theta \, \mathrm{d}V - \int_{\partial\Omega_q} q_n \delta\theta \, \mathrm{d}A = 0 \tag{99}$$

$$\int_{\Omega} \omega'(d) \, \mathcal{H} \delta d \, \mathrm{d}V + \int_{\Omega} \frac{G_c \alpha'}{c_0 l} \delta d \, \mathrm{d}V + \int_{\Omega} \frac{2 G_c l}{c_0} \nabla d \cdot \nabla \delta d \, \mathrm{d}V + \int_{\Omega} \eta \dot{d} \delta d \, \mathrm{d}V = 0 \tag{100}$$

$$\int_{\Omega} \dot{c} \delta c \, \mathrm{d}V + \int_{\Omega} D \nabla c \cdot \nabla \delta c \, \mathrm{d}V - \int_{\Omega} M \frac{\mathrm{d}n}{\mathrm{d}t} \delta c \, \mathrm{d}V - \int_{\partial\Omega_n} J_n \delta c \, \mathrm{d}A = 0 \tag{101}$$

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